

$$w = \sum_{i=1}^m \alpha_i y^{(i)} x^{(i)} \rightarrow \phi(x^{(i)})$$

COL 774

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$$\phi(x^{(i)})^T \cdot \phi(x^{(j)})$$

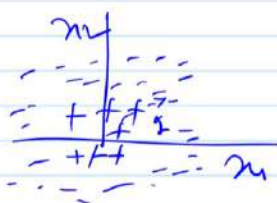
SVM Dual: $\max_{\alpha} \sum_{i=1}^m \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j [x^{(i)T} x^{(j)}] y^{(i)} y^{(j)}$

SMO sequential minimal optimization: $0 \leq \alpha_i \leq C$ $\sum_i \alpha_i y^{(i)} = 0$

Kernel Function

Linear SVMs

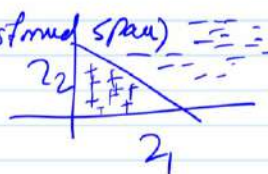
Non-Linear SVMs



separable $x_1^2 + x_2^2 \leq 1$

$z_1 = x_1^2$ $z_2 = x_2^2$

$z_1 + z_2 \leq 1$ (Transformed space)



$\phi: \mathbb{R}^n \rightarrow \mathbb{R}^N$

ϕ (degree 2 transformation)

Kernel Function:-

$K: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$

$K(x, z) \equiv \phi(x)^T \phi(z)$

$x \in \mathbb{R}^2$

$$\phi(x) = \begin{bmatrix} x_1^2 \\ x_2^2 \\ x_1 x_2 \\ \sqrt{c} x_1 \\ \sqrt{c} x_2 \\ c \end{bmatrix}$$

$$\phi(z) = \begin{bmatrix} z_1^2 \\ z_2^2 \\ z_1 z_2 \\ \sqrt{c} z_1 \\ \sqrt{c} z_2 \\ c \end{bmatrix}$$

$\phi(x)^T \phi(z) = ?$

$(x^T z + c)^2$

$(x^T z)^2 + 2x^T z c + c^2$

$= \left[\sum_i x_i z_i \right]^2 + \sum_i \sqrt{c} x_i z_i \sqrt{c} + c c$

$(x_1 z_1 + x_2 z_2)^2 = x_1^2 z_1^2 + x_2^2 z_2^2 + x_1 z_1 x_2 z_2 + x_2 z_2 x_1 z_1$

n : - variables

degree: - d

$\rightarrow$ space of $\phi(x)$

$0 \mid 0 \mid 0 \mid 0 \mid 0 \mid 0 \mid 0 \mid 0$

$n+d \mid d \mid \overline{N}$

x_1, x_2, x_3, x_4, x_5

$0 \mid 0 \mid 0 \mid 0 \mid 0 \mid 0 \mid 0 \mid 0$
 x_1^2, x_2, x_3

x_1^d
 $x_1^{d-1} x_2$
 $x_1^{d-2} x_2^2$
 $\vdots$
 $x_1 x_2^{d-1}$
 x_2^d

$$\phi(x)^T \phi(z) \stackrel{?}{=} \underbrace{(a^T z + c)^d}_{\text{(for a degree } d \text{ transformation)}} = K(x, z)$$

$$\Rightarrow \text{Kernel Function} = \left. \begin{array}{l} K: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R} \text{ s.t. } \exists \phi: \mathbb{R}^n \rightarrow \mathbb{R}^N \\ \text{s.t. } K(x, z) = \phi(x)^T \phi(z) \\ x, z \in \mathbb{R}^n \end{array} \right\}$$

① Polynomial kernel: $K(x, z) = (a^T z + c)^d$

② Gaussian Kernel:- $K(x, z) = e^{-\frac{\|x - z\|^2}{2\sigma^2}}$

Merccor's Theorem:-

Let K :- kernel &

$x^{(1)} \dots x^{(m)}$ be a set of points $x^{(i)} \in \mathbb{R}^n$ ($i \in \{1, \dots, m\}$)

then:- K is a valid kernel

if & (a) K^M is symmetric

(b) K^M is semi-definite
 $\forall \{x^{(i)}\}_{i=1}^m$ finite

Kernel matrix:- K :- kernel

$x^{(1)} \dots x^{(m)} \quad K: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$

$$K^M \equiv \begin{bmatrix} K(x^{(1)}, x^{(1)}) & \dots & K(x^{(1)}, x^{(m)}) \\ \vdots & \ddots & \vdots \\ K(x^{(m)}, x^{(1)}) & \dots & K(x^{(m)}, x^{(m)}) \end{bmatrix}$$

$$K_{ij}^M \equiv K(x^{(i)}, x^{(j)})$$

if $K(x, z) = \phi(x)^T \phi(z)$

then K^M is (a) + the semi-definite
 (b) symmetric

$$u^T K^M u \geq 0 = \sum_{i,j} u_i K(x^{(i)}, x^{(j)}) u_j$$

$$= \sum_{i,j} u_i \phi(x^{(i)})^T \phi(x^{(j)}) u_j \cdot K_{ij}^M$$

$$= \sum_{i,j} u_i \left[\sum_k \phi(x^{(i)})_k \phi(x^{(j)})_k \right] u_j$$

$$= \sum_{i,j} \sum_k \left[u_i \phi(x^{(i)})_k \quad u_j \phi(x^{(j)})_k \right]$$

$$= \sum_k \left[\sum_i u_i \phi(x^{(i)})_k \right] \left[\sum_j u_j \phi(x^{(j)})_k \right] = \sum_k \left(\sum_i u_i \phi(x^{(i)})_k \right)^2 \geq 0$$